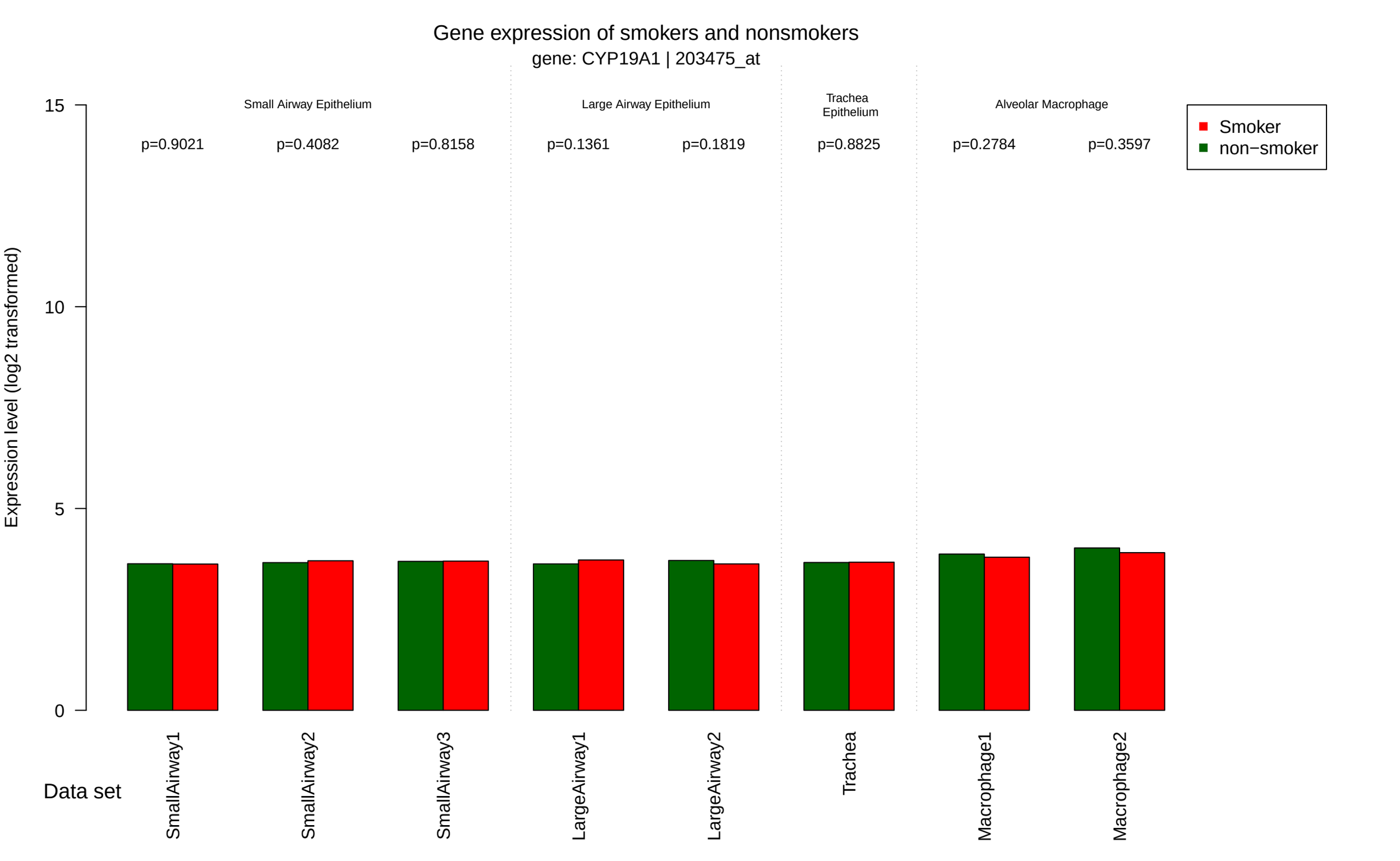
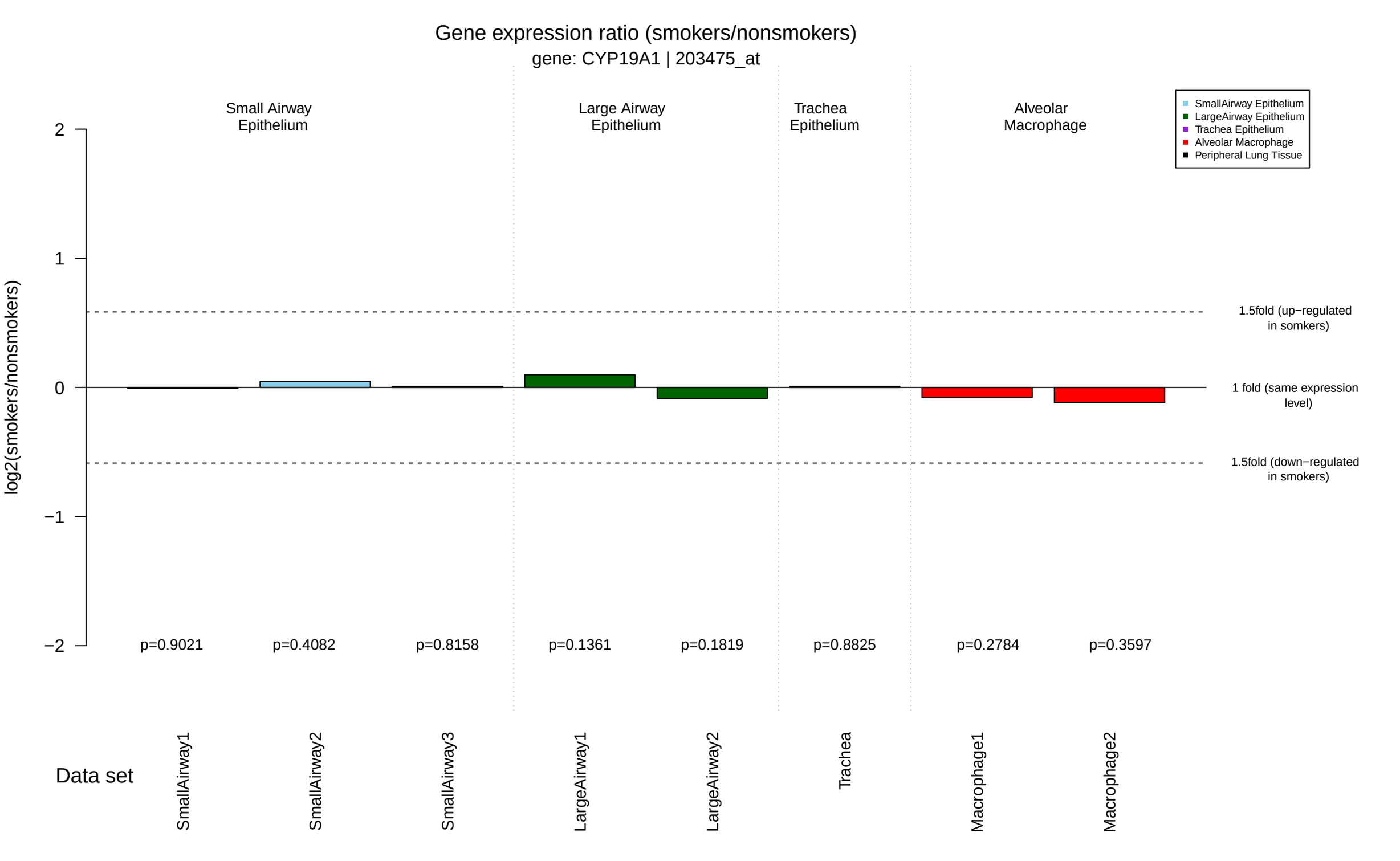
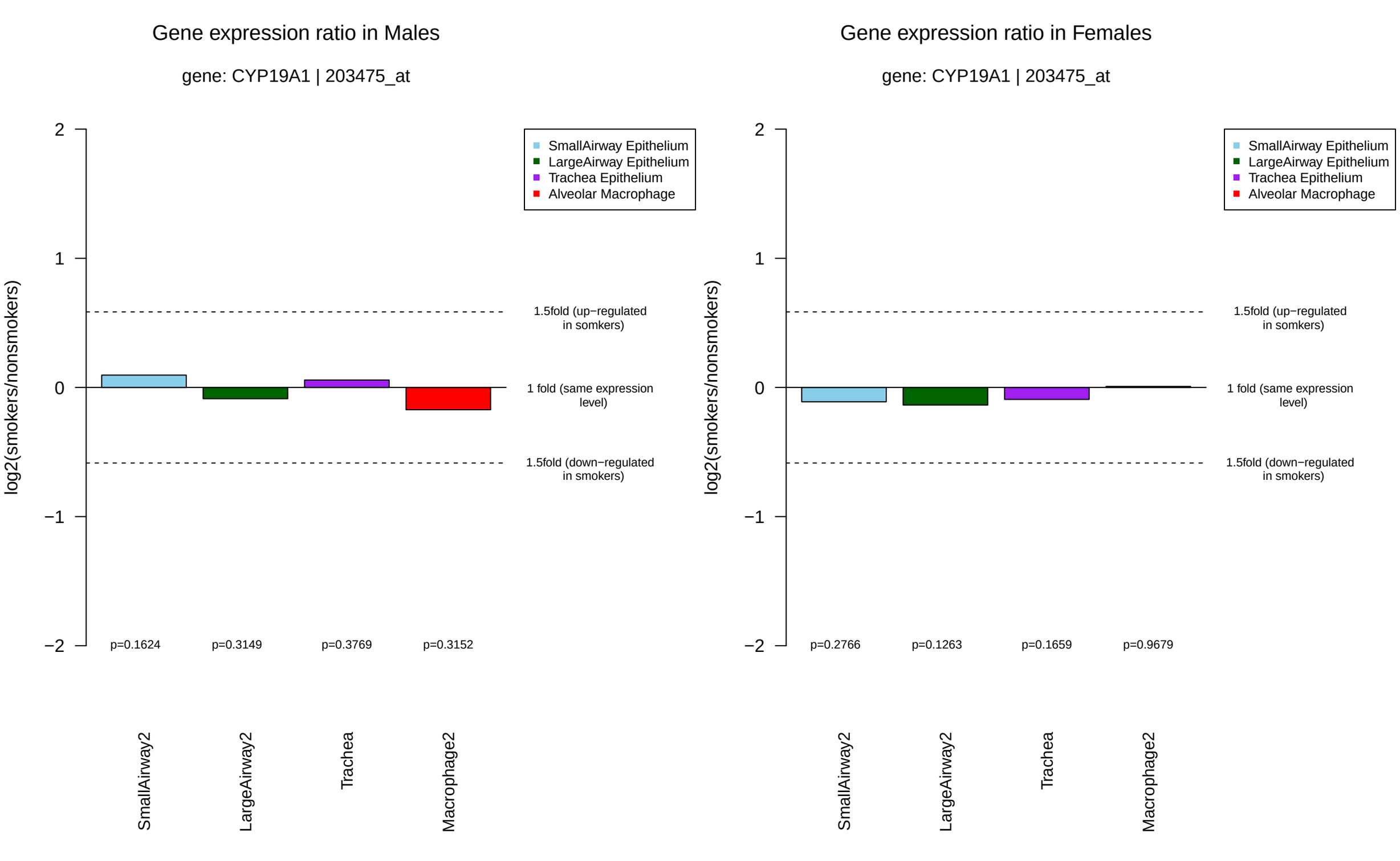
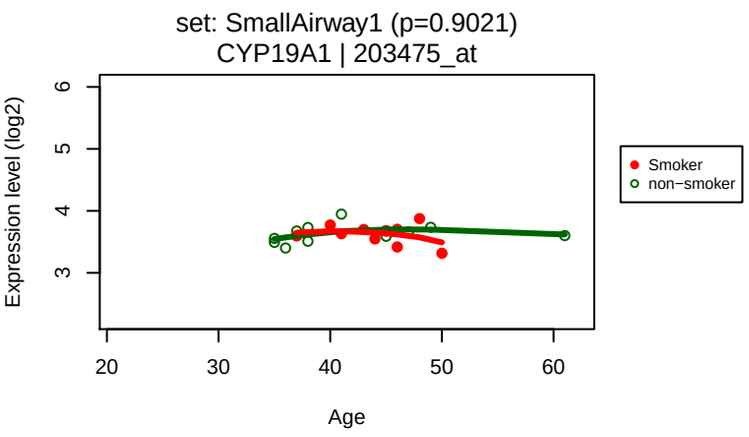
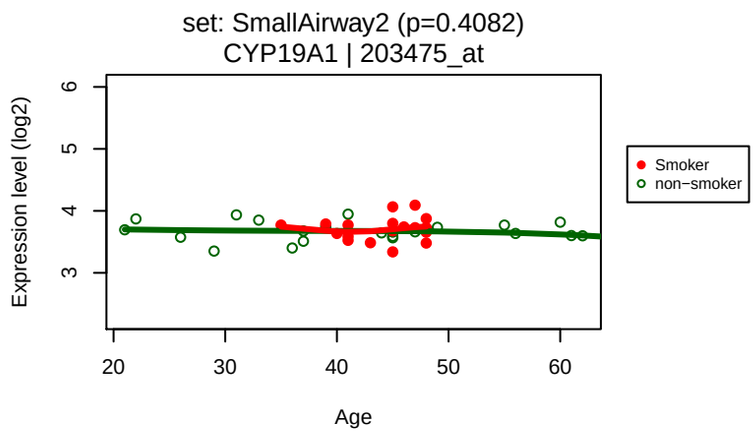
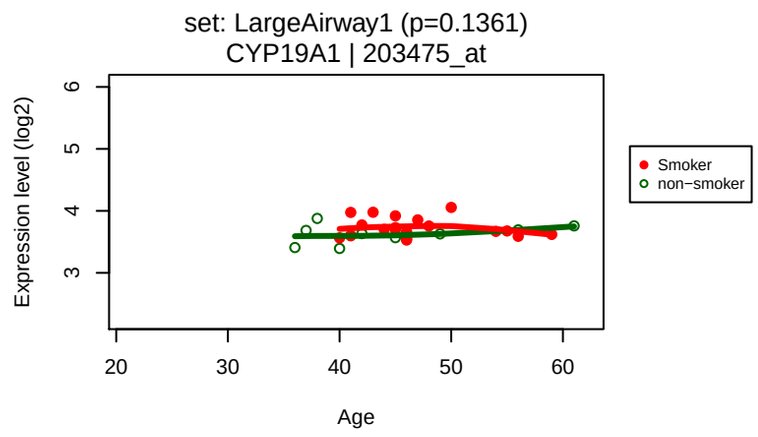

Gene expression of smokers and nonsmokers
gene: CYP19A1 | 203475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9021
p=0.4082
p=0.8158
p=0.1361
p=0.1819
p=0.8825
p=0.2784
p=0.3597
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

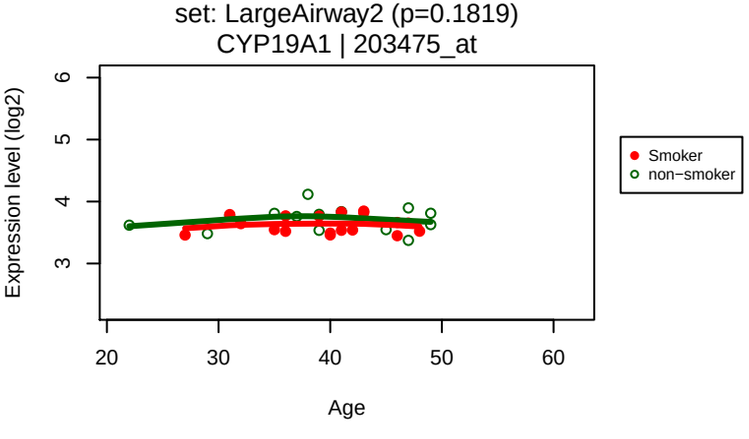

Gene expression ratio (smokers/nonsmokers)
gene: CYP19A1 | 203475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9021
p=0.4082
p=0.8158
p=0.1361
p=0.1819
p=0.8825
p=0.2784
p=0.3597
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

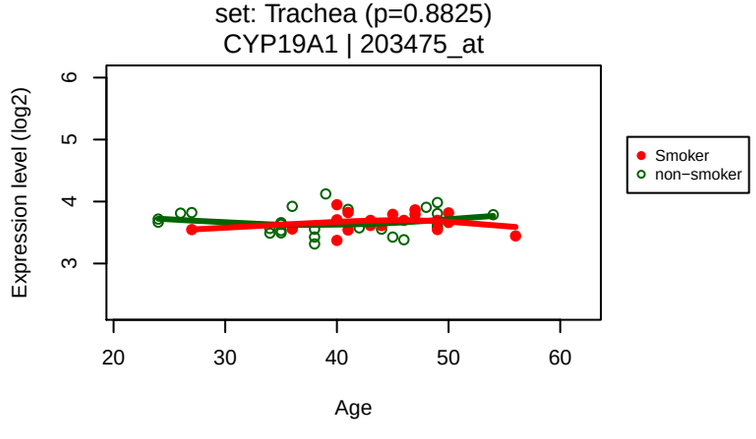

Gene expression ratio in Males
gene: CYP19A1 | 203475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1624
p=0.3149
p=0.3769
p=0.3152
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP19A1 | 203475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2766
p=0.1263
p=0.1659
p=0.9679
SmallAirway2
LargeAirway2
Trachea
Macrophage2

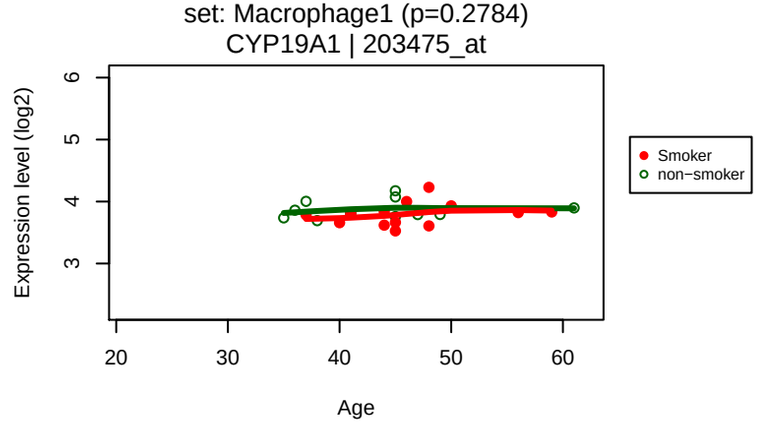

set: SmallAirway1 (p=0.9021)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker

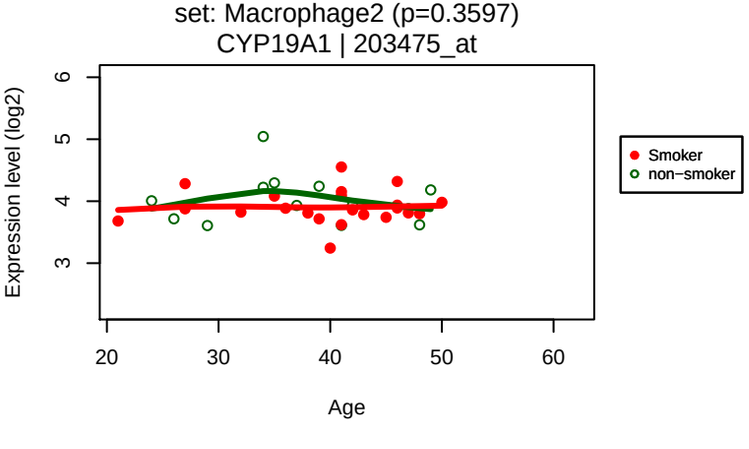

set: SmallAirway2 (p=0.4082)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker

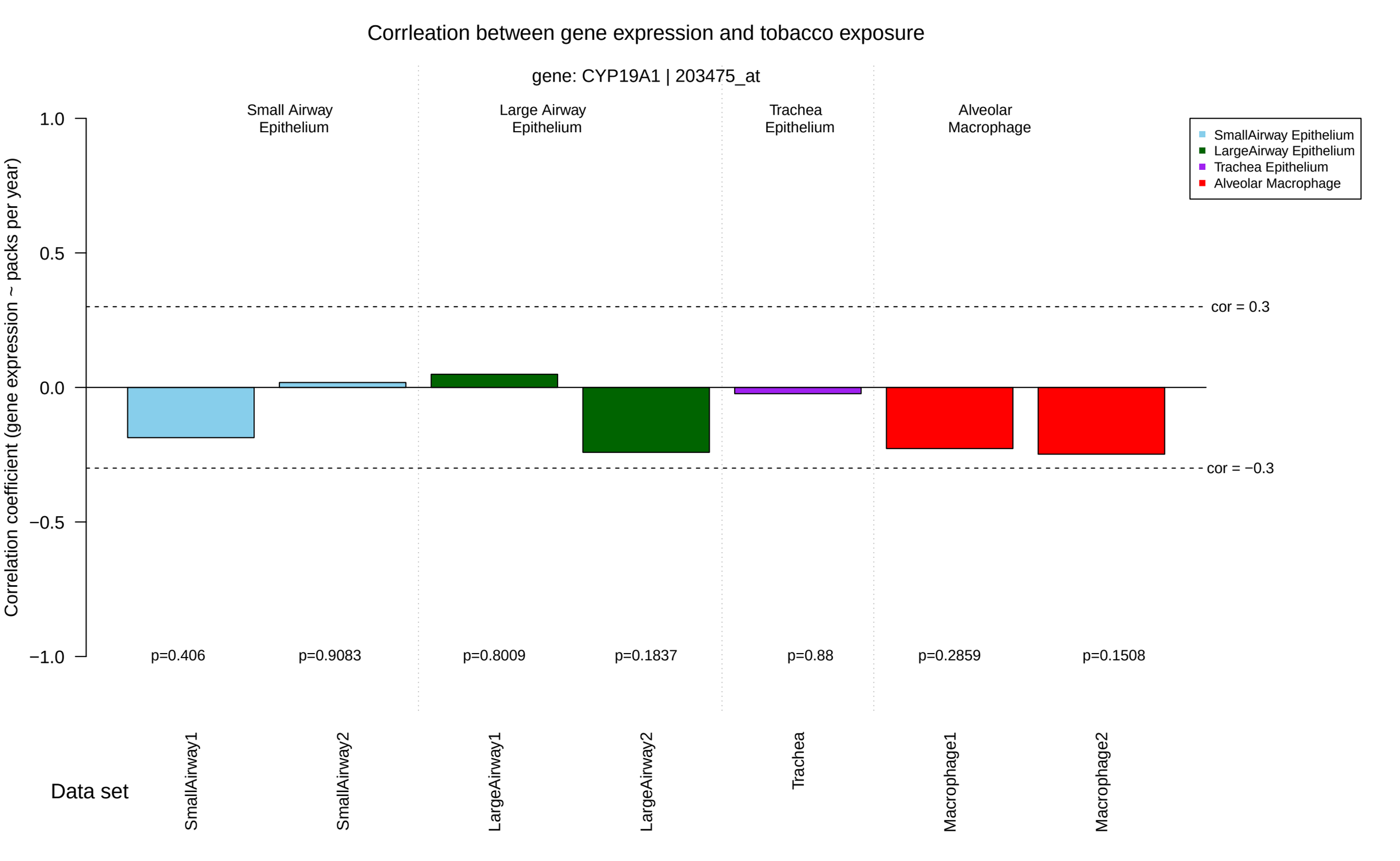

set: LargeAirway1 (p=0.1361)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1819)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8825)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2784)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3597)
CYP19A1 | 203475_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP19A1 | 203475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.406
p=0.9083
p=0.8009
p=0.1837
p=0.88
p=0.2859
p=0.1508
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2